Sudarshan S Harithas

Ph.D. Candidate in Computer Science, Brown University

sudarshan_harithas@brown.edu | sudarshan-s-harithas.github.io | Google Scholar

Education

2023–Present **Brown University**
Ph.D. Candidate in Computer Science.

2021–2023 **International Institute of Information Technology, Hyderabad (IIIT-H)**
M.S. by Research in Computer Science and Engineering.
Thesis: "From Scene Representation to Robust Autonomy: A Seamless Integration."

2016–2020 **B.M.S. College of Engineering**
B.E. in Electronics and Communication Engineering.

Research Publications

Conference Publications

1. **BRACE: Adapting Whole-Body References for Force and Terrain Aware Humanoid Motion Tracking**
Sudarshan Harithas, Chen Yu, Juan Borbon, Shubhankar Mondal, Winston Zha, Srinath Sridhar, Dingqi Zhang, Jiuguang Wang.
Preprint, 2026; under review. Paper Project page
2. **Deliberate Practice: Learning Robot Skills under a Budget**
Shivam Vats, **Sudarshan Harithas**, Mete Tuluhan Akbulut, Arvind Raghunathan, George Konidaris.
Preprint, 2026; under review. Paper
3. **DreamControl-v2: Simpler and Scalable Autonomous Humanoid Skills via Trainable Guided Diffusion Priors**
Sudarshan Harithas, Sangkyung Kwak, Pushkal Katara, Srujan Deolasee, Dvij Kalaria, Srinath Sridhar, Sai Vemprala, Ashish Kapoor, Jonathan Chung-Kuan Huang.
IEEE/RSJ IROS 2026. Paper Project page
4. **DreamControl: Human-Inspired Whole-Body Humanoid Control for Scene Interaction via Guided Diffusion**
Dvij Kalaria, **Sudarshan Harithas**, Pushkal Katara, Sangkyung Kwak, Sarthak Bhagat, S. Shankar Sastry, Srinath Sridhar, Sai Vemprala, Ashish Kapoor, Jonathan Huang.
IEEE ICRA 2026. **[Oral presentation]** Paper Project page
5. **MotionGlot: A Multi-Embodied Motion Generation Model**
Sudarshan S Harithas, Srinath Sridhar.
IEEE ICRA 2025; also presented at the CoRL 2024 X-Embodiment Workshop. Paper Project page Video
6. **λ: A Benchmark for Data-Efficiency in Long-Horizon Indoor Mobile Manipulation Robotics**
Ahmed Jaafar, Shreyas Sundara Raman, **Sudarshan Harithas**, Yichen Wei, Sofia Juliani, Anneke Wernerfelt, Benedict Quartey, Ifrah Idrees, Jason Xinyu Liu, Stefanie Tellex.
IEEE/RSJ IROS 2025. Paper Project page
7. **FinderNet: A Data Augmentation Free Canonicalization Aided Loop Detection and Closure Technique for Point Clouds in 6-DOF Separation**
Sudarshan S Harithas, Gurkirat Singh, Aneesh Chavan, Sarthak Sharma, Suraj Patni, Chetan Arora, K. Madhava Krishna.
IEEE/CVF WACV 2024. **[Round 1 acceptance]** Paper Project page Video
8. **CCO-VOXEL: Chance Constrained Optimization over Uncertain Voxel-Grid Representation for Safe Trajectory Planning**
Sudarshan S Harithas, Rishabh Dev Yadav, Deepak Singh, Arun Kumar Singh, K. Madhava Krishna.
IEEE ICRA 2022. Paper Code Video
9. **UrbanFly: Uncertainty-Aware Planning for Navigation Amongst High-Rises with Monocular Visual-Inertial SLAM Maps**
Sudarshan S Harithas, Ayyappa Thatavarthy, Gurkirat Singh, Arun Kumar Singh, K. Madhava Krishna.
American Control Conference (ACC) 2023. Paper Code Video
10. **RP-VIO: Robust Plane-Based Visual-Inertial Odometry for Dynamic Environments**
Karnik Ram, Chaitanya Kharyal, **Sudarshan S Harithas**, K. Madhava Krishna.
IEEE/RSJ IROS 2021. Paper Project page Video

Preprint

ARDOP: A Versatile Humanoid Robotic Research Platform
Sudarshan S Harithas, Harish V. Mekali.
arXiv preprint, 2021. Paper Project page

Work Experience

Multiply Labs — Research Intern
June–August 2026 — Advisor: Dr. Jiuguang Wang

- Developed force- and terrain-aware whole-body humanoid controllers for BRACE, adapting motion references to external loads and sloped terrain (under review).

General Robotics — Research Intern
June–August 2025 — Advisor: Dr. Jonathan Huang

- Developed robot-space diffusion models for whole-body humanoid locomotion and manipulation skills (ICRA 2026; IROS 2026).
- Parallelized human-to-humanoid motion retargeting on GPUs and demonstrated scaling laws for skill transfer (ICRA 2026; IROS 2026).

Interactive 3D Vision & Learning Lab, Brown University — Graduate Research Assistant
August 2023–Present — Advisor: Prof. Srinath Sridhar

- Developed cross-embodied motion-generation policies spanning human and quadruped embodiments (ICRA 2025; CoRL 2024 X-Embodiment Workshop).
- Worked on sim-to-real reinforcement learning for whole-body humanoid control, including force- and terrain-aware motion tracking (ICRA 2026; IROS 2026; under review).

Robotics Research Center, IIIT Hyderabad — Advisor: Prof. K. Madhava Krishna

> **Graduate Research Assistant** — August 2021–July 2023
> - Developed FinderNet, a robust 6-DOF loop detection and closure system for point clouds (WACV 2024).
> - Developed real-time uncertainty-aware motion planning for quadrotors (ICRA 2022; ACC 2023).
>
> **Research Intern** — August 2020–July 2021
> - Worked on plane-based visual-inertial odometry in highly dynamic environments (IROS 2021).

Summer Geometry Initiative, MIT — Research Fellow
July–August 2022 — Organizer: Prof. Justin Solomon

- Implemented DeltaConv for anisotropic point-cloud convolutions and developed a fast surface triangulation algorithm for NeuralODF.

Awards and Grants

June 2025	**NVIDIA Academic Grant Program** "Large Action Models: A Cross-Embodied Model for Dexterous Manipulation." Co-written with Srinath Sridhar and Tao Lu.
May 2025	**Outstanding Reviewer, CVPR 2025**
April 2025	**Doctoral Consortium, ICRA 2025** **[Oral Highlight]**
2021–2023	**Research Fellowship, IIIT Hyderabad**
March 2022	**IEEE RAS Travel Grant, ICRA 2022**

Academic Service

August 2026	**Co-organizer, NeuRo-SymBolic World Models (RoBoWoMo) Workshop**
January 2026	**CS Ph.D. Application Reader, Brown University**
Reviewer	**CoRL, ICRA, IROS, CVPR, ECCV, and RA-L**

Invited Talks

October 2026	**BRACE: Adapting Whole-Body References for Force and Terrain Aware Humanoid Motion Tracking** General Robotics.
April 2025	**Large Action Models: Towards Building a Foundation Model for Cross-Embodied Action Generation** Robotics and AI Institute (RAI).

Media Coverage

MotionGlot: Brown University News (May 8, 2025) | The Boston Globe (December 15, 2025)